

Digital Assignment - 8

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Register Number: 23MID0078

Course Name: MDI3003 - Advanced Predictive Analytics (Lab)

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GitHub Repository:
https://github.com/raghavmri/23MID0078-advanced-predictive-analytics

Agricultural Predictive Analytics

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1) Aim and Objectives

Aim

The primary aim of this laboratory investigation is to design, construct, audit, and benchmark a reproducible predictive regression pipeline for district-level rice yield forecasting across India under a rigorous prospective evaluation protocol. A companion guided classification pipeline evaluates environmental and soil-nutrient feature suitability for multiclass crop recommendation.

Objectives

- **Prediction Contract:** Establish a formal problem definition specifying the observation unit, continuous target metric (t/ha), operational forecast horizon, and strict pre-harvest predictor availability to prevent information leakage.
- **API Data Acquisition and Auditing:** Ingest the ICRISAT panel dataset via its public REST API, verifying checksums and variable definitions, followed by automated ingestion of the environmental crop recommendation dataset.
- **Chronological Partitioning:** Implement strict temporal splitting reserving the most recent observed years as a locked test set and preceding years for validation, ensuring historical estimation is never contaminated by future data.
- **Leakage-Free Preprocessing:** Encapsulate categorical encodings and continuous feature standardizations within a pipeline fitted exclusively on training instances.

2) Dataset Description and Quality Analysis

Data Provenance and Schema

The core regression experiment uses Dataset D5, a versioned agricultural research repository by Souryabrata Mohapatra: *ICRISAT District-Level Data: Heterogeneous Climate Effect on Crop Yield and Associated Risks to Water Security in India*. The dataset harmonizes ICRISAT Meso-level data with meteorological observations across twenty Indian states. Rather than manual downloads, the raw tabular file was retrieved via the Mendeley Data Public REST API. The panel spans 1990 to 2015, capturing 560 districts across 12,803 raw survey rows.

Schema Transformation and Unit Standardization

- **Unit Harmonization:** Rice yield was scaled from kg/ha to metric tonnes per hectare (t/ha) by dividing by 1000.0, establishing yield_t_ha.
- **Target Reconciliation:** Gross production divided by cultivated area was cross-checked against reported yield; the median absolute deviation was less than 0.0003 t/ha, confirming discrepancies were purely rounding truncation.
- **Scope Restriction:** The study is strictly bounded to rice (*Oryza sativa*), India's primary staple crop, avoiding inter-crop variance and packing ambiguities.
- **Key Uniqueness:** Auditing the composite key (state, district, season, year) revealed zero duplicated records. Excluding 52 upstream records with missing targets yielded a canonical dataset of 12,751 complete district-harvest records.

Information Leakage and Predictor Whitelist

Post-harvest variables (gross production, harvested area) and full-season rainfall totals are unavailable prior to harvest. including them enables exact mathematical reconstruction of yield, creating severe data leakage. The predictor space was restricted to a strict pre-harvest whitelist:

- state: Administrative state name (nominal categorical string).
- district: Administrative district identifier (nominal categorical string, capturing localized agro-ecological micro-climates).
- season: Crop season (standardized to "Annual").
- year: Integer harvest year index (temporal coordinate for secular technological trends).

Classification Extension Dataset

The Atharva Ingle Crop Recommendation Dataset (Dataset D3) comprises 2,200 observations across seven physical, chemical, and climatic features: Nitrogen (N, kg/ha), Phosphorus (P, kg/ha), Potassium (K, kg/ha), Temperature (°C), Relative Humidity (%), Soil pH (0–14), and Rainfall (mm). The target comprises 22 balanced crop classes (100 samples each). Quality screening confirmed zero missing, infinite, or duplicate entries.

3) Exploratory Data Analysis

Training Yield Distribution Analysis

Exploratory analysis was restricted exclusively to training data (1990–2011) to preserve validation and test set independence.

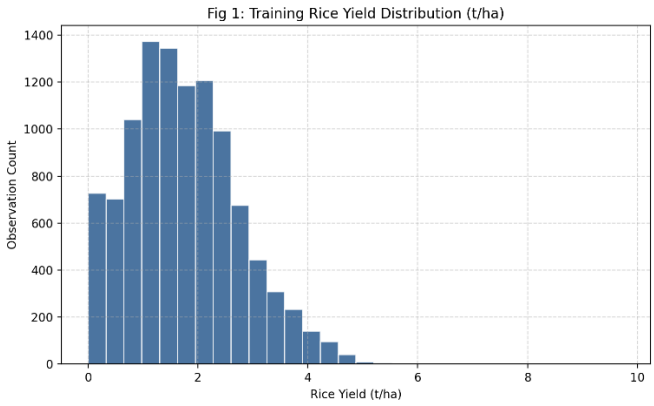


Fig 1

Interpretation: Fig. 1 shows a unimodal, moderately right-skewed yield distribution, with a mean of 1.7358 t/ha. About 7.4% of observations record zero yield, while the tail reaches 9.7500 t/ha, reflecting high-yield irrigated regions.

Temporal Observation Coverage and Geographic Balance

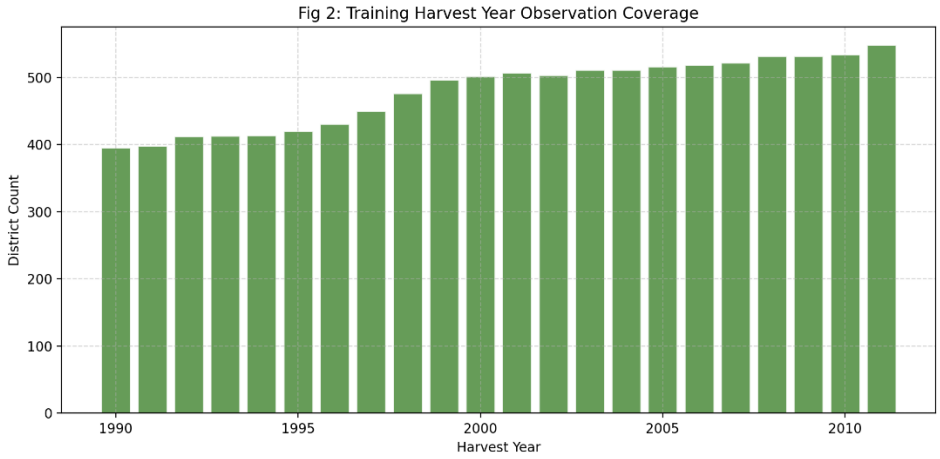


Fig 2

Interpretation: Fig. 2 shows stable district-level coverage across 22 training years, increasing from 400–520 active districts in the 1990s to 545 by 2011, reflecting improved administrative and statistical reporting.

Environmental Feature Profiles in Crop Recommendation

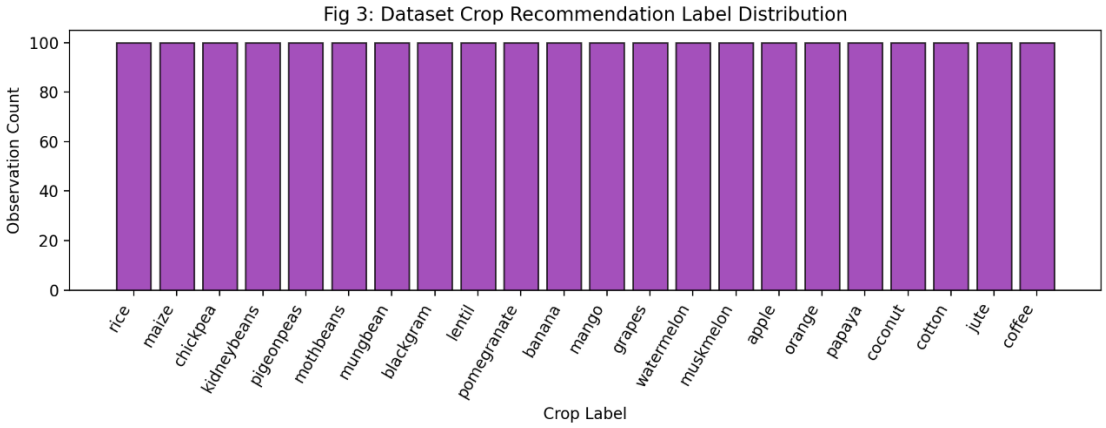


Fig 3

Interpretation: Fig. 3 shows a uniform and balanced distribution across all 22 target crop varieties, with 100 instances per variety.

Fig 4: Training Soil pH Distribution

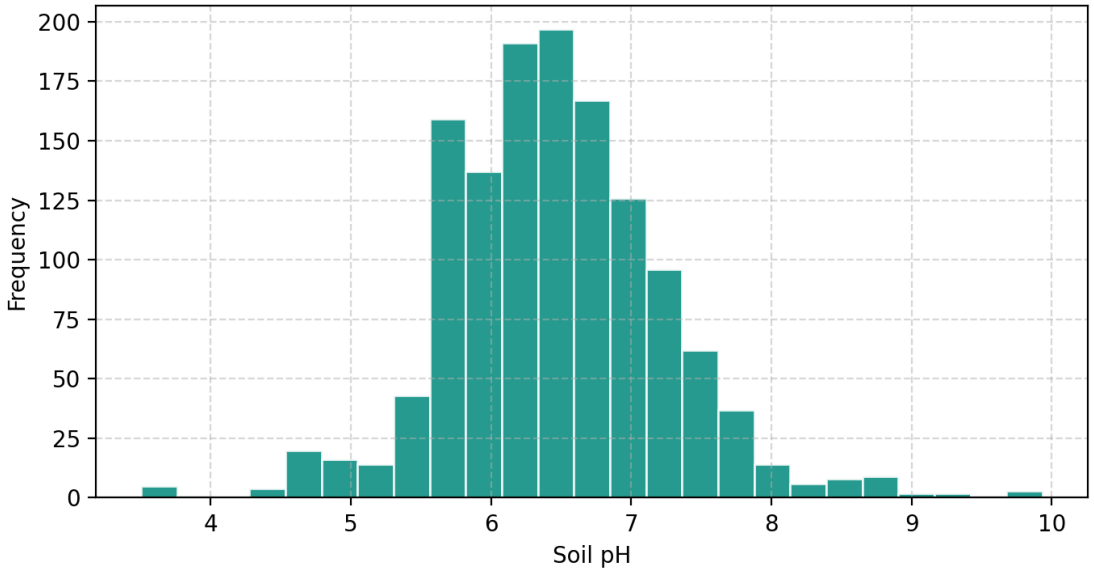


Fig 4

Interpretation: Fig. 4 shows a near-Gaussian soil pH distribution centered at 6.47, spanning acidic to moderately alkaline conditions.

4) Data Preprocessing and Train-Test Split

Prospective Chronological Partitioning

Standard random cross-validation allows models to learn future harvest conditions when predicting past yields. To replicate operational prospective forecasting, a chronological holdout partition was enforced:

- **Test Partition (Locked):** Harvest years 2014–2015 (final 2 observed years), with 1,114 observations across 552 unique districts.
- **Validation Partition:** Harvest years 2012–2013 (intermediate 2 years), with 1,096 observations across 544 unique districts.

- **Training Partition:** Harvest years 1990–2011 (historical 22 years), with 10,541 observations across 545 unique districts.

Partition	Harvest Years	Observations	Districts	States	Mean Yield (t/ha)	SD (t/ha)	Min (t/ha)	Max (t/ha)
Training	1990–2011	10,541	545	20	1.7358	1.0104	0.0000	9.7500
Validation	2012–2013	1,096	544	20	2.1922	1.1138	0.0000	6.6400
Locked Test	2014–2015	1,114	552	20	2.1212	1.1893	0.0000	5.8560

→ Mean yield rises from 1.7358 t/ha in training to 2.1922 t/ha in validation and 2.1212 t/ha in test. This positive secular shift reflects hybrid cultivar adoption, rural electrification, and improved agronomic practices over the quarter-century window.

Leakage-Free Pipeline Construction

Transformers were encapsulated within a ColumnTransformer fitted strictly on training data:

- **Categorical Pipeline:** state, district, and season were encoded via OneHotEncoder(handle_unknown='ignore'), ensuring novel districts encountered in prospective evaluation map to zero-vectors without errors or bias.
- **Numeric Pipeline:** year was processed via SimpleImputer(strategy='median') and scaled to zero mean and unit variance using StandardScaler. Standardizing year balances L2 regularization scales in Ridge while leaving tree splits invariant.

- **Isolation Audit:** Programmatic assertions confirmed that the categorical encoder registered exactly the 545 training districts and 20 training states, with zero post-2011 categories leaking into the pipeline.

5) Model Implementation

Model Implementation

Four regression models were evaluated for yield prediction:

Model	Key Configuration
Median Baseline	Training-set median
Ridge Linear Trend	$\alpha = 1.0$
Decision Tree	Maximum depth = 6
Random Forest	60 trees, maximum depth = 12

For crop recommendation, three classification models were evaluated:

Model	Configuration
Majority Baseline	Most frequent crop class
Logistic Regression	Linear multiclass classifier
Random Forest	Tree-based multiclass classifier

6) Model Evaluation and Comparison

Validation Protocol and Selection Metric

Model selection was governed strictly by prospective performance on the 2012–2013 validation set. Mean Absolute Error (MAE) was designated as the primary selection criterion as it directly reflects operational errors in tonnes per hectare without over-penalizing isolated weather shocks. Root Mean Squared Error (RMSE) and R^2 serve as secondary diagnostics. Mean Absolute Percentage Error (MAPE) was omitted because zero-yield records produce division-by-zero errors.

Validation Results

Model Architecture	Validation MAE (t/ha)	Validation RMSE (t/ha)	Validation R^2	Fit Time (s)	Model Selection Status
Ridge Linear Trend	**0.4770**	**0.6452**	**0.6641**	**0.0327**	**Selected (Best Candidate)**
Random Forest (Depth 12)	0.6196	0.7874	0.4998	0.4369	Rejected
Random Forest (Depth 6)	0.7555	0.9258	0.3084	0.2278	Rejected
Decision Tree (Depth 6)	0.7727	0.9523	0.2683	0.0203	Rejected
Median Constant Baseline	0.9816	1.2391	-0.2389	0.0185	Rejected (Benchmark)

Methodological Analysis of Model Performance

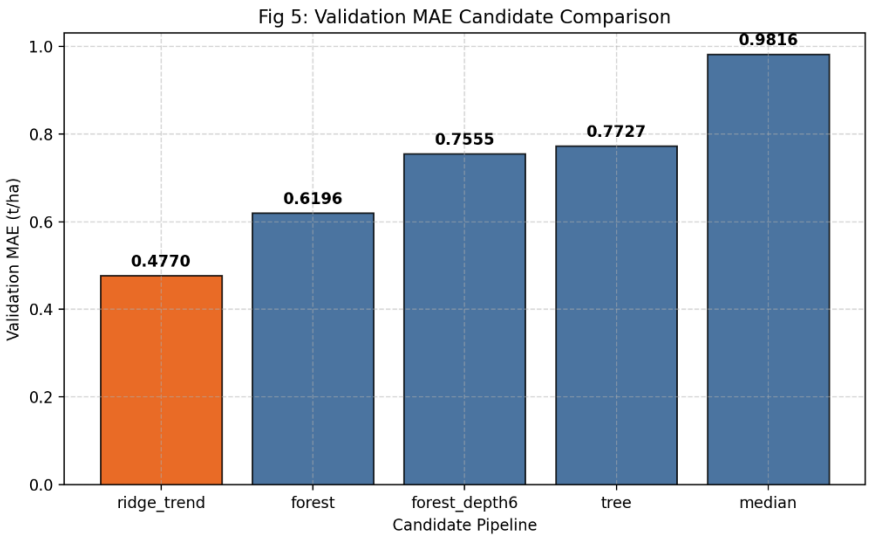


Fig 5

Interpretation: Fig. 5 shows Ridge achieving the lowest validation error (MAE: 0.4770 t/ha), outperforming Random Forest, Decision Tree, and the median baseline. Its linear time trend better captures rising yields and avoids the underprediction seen in tree-based models.

7) Feature and Preprocessing Ablation Analysis

Tree Depth Ablation

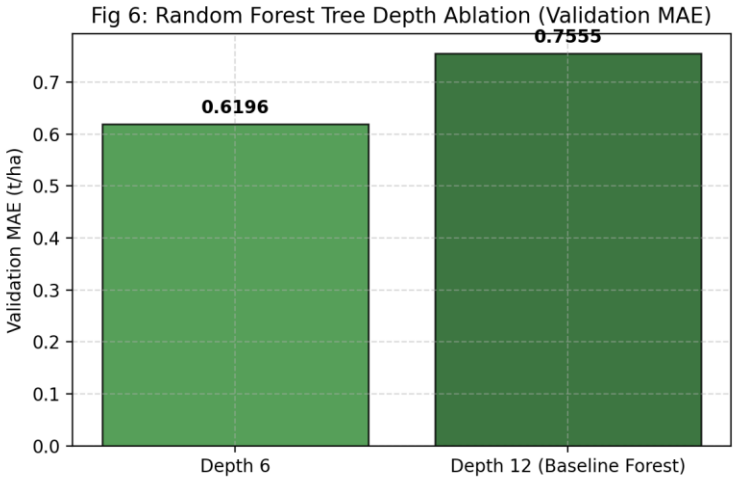


Fig 6

Interpretation: Fig. 6 shows that limiting Random Forest depth to 6 worsens MAE from 0.6196 to 0.7555 t/ha and reduces R^2 from 0.4998 to 0.3084. Depth 12 provides better regional partitioning while maintaining controlled variance.

Spatial Feature Ablation (District Granularity)

To quantify the predictive value of administrative boundaries, an ablation experiment removed district, retaining only state, season, and year.

Model Architecture	Full Features MAE (t/ha)	Omitted District MAE (t/ha)	Absolute Degradation (t/ha)	Relative Degradation (%)
Ridge Linear Trend	0.4770	0.6668	+0.1898	+39.79%
Random Forest	0.6196	0.6665	+0.0470	+7.58%

→ Omitting district identifiers degraded Ridge accuracy by 39.79%. Agricultural productivity is intensely localized; aggregating spatial variance to state levels discards critical agro-climatic disparities between semi-arid rain-fed districts and canal-irrigated basins within the same state.

Prospective Rolling-Origin Temporal Cross-Validation

To audit model temporal stability, rolling-origin backtesting was executed across five historical cutoffs (evaluation years 2009–2013). At each origin T, models were trained strictly on data prior to T and evaluated prospectively on year T.

Cutoff Year (T)	Training Window	Sample Size (N)	Median Baseline MAE (t/ha)	Ridge Trend MAE (t/ha)	Random Forest MAE (t/ha)
2009	1990–2008	532	0.8996 (R^2 = -0.0297)	**0.3313** (R^2 = 0.8456)	0.4396 (R^2 = 0.7513)

Cutoff Year (T)	Training Window	Sample Size (N)	Median Baseline MAE (t/ha)	Ridge Trend MAE (t/ha)	Random Forest MAE (t/ha)
2010	1990–2009	534	0.9011 (R^2 = -0.0706)	**0.3753** (R^2 = 0.7254)	0.4891 (R^2 = 0.6093)
2011	1990–2010	548	0.9435 (R^2 = -0.2023)	**0.4445** (R^2 = 0.6685)	0.5792 (R^2 = 0.4974)
2012	1990–2011	548	0.9585 (R^2 = -0.2573)	**0.4820** (R^2 = 0.6091)	0.6440 (R^2 = 0.3799)
2013	1990–2012	548	0.9919 (R^2 = -0.1982)	**0.4438** (R^2 = 0.7354)	0.5713 (R^2 = 0.6081)

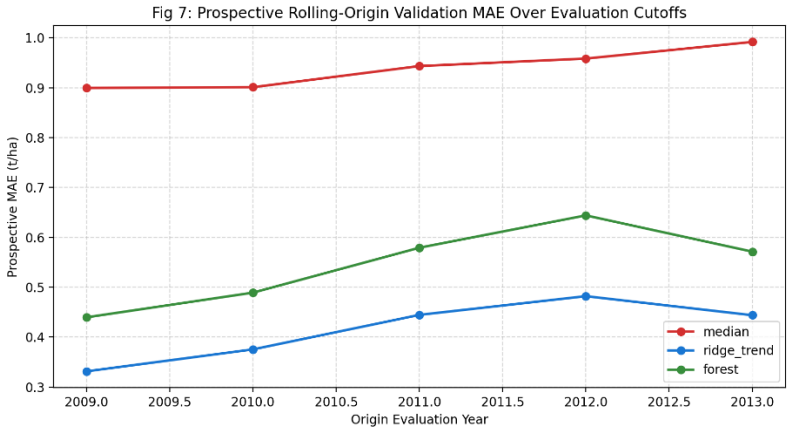


Fig 7

Interpretation: Fig. 7 shows Ridge consistently achieving lower prospective MAE than Random Forest and the Median Baseline across all five evaluation years, maintaining its superiority despite annual monsoon-driven variability.

8) Test Evaluation and Diagnostic Analysis

Single-Pass Locked Test Scoring

Following model selection, the test partition (years 2014–2015, N = 1,114) was evaluated exactly once. A TEST_LOCK token was logged prior to scoring, and hyperparameters were held fixed without post-hoc modification.

Model Pipeline	Test MAE (t/ha)	Test RMSE (t/ha)	Test R^2	Inference Latency (N=1,114)
Ridge Linear Trend (Selected)	**0.4957**	**0.6630**	**0.6889**	0.0065 s
Median Baseline (Benchmark)	1.0195	1.2794	-0.1584	0.0001 s

→ The selected Ridge pipeline achieved an MAE of 0.4957 t/ha on unseen test data, closely matching its validation MAE (0.4770 t/ha) and demonstrating generalization without overfitting. The median baseline produced a test MAE of 1.0195 t/ha and negative R^2 (-0.1584), confirming the failure of static historical averages under secular drift.

Year-by-Year Robustness Analysis

Disaggregating performance across test years confirmed consistent accuracy across prospective seasons:

Harvest Year	Sample Size (N)	MAE (t/ha)	RMSE (t/ha)	R^2
2014	557 districts	0.5247	0.7024	0.6443
2015	557 districts	0.4667	0.6212	0.7267

Diagnostic Visualizations

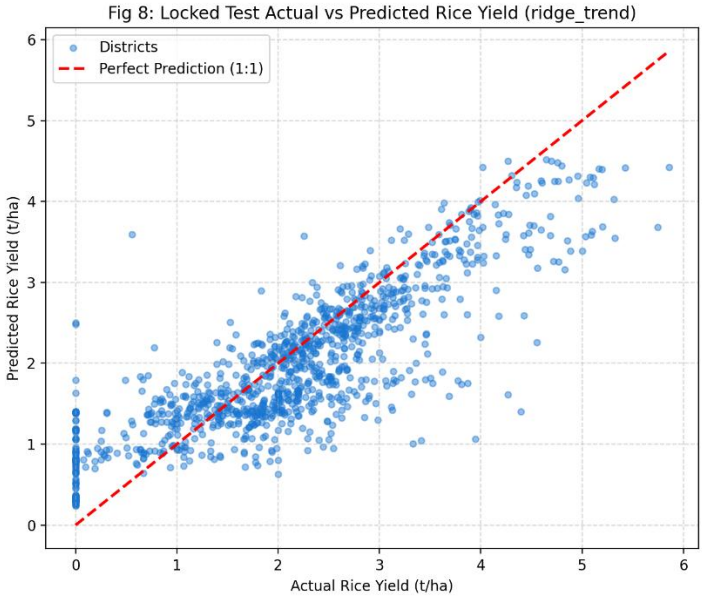


Fig 8: Locked Test Actual vs Predicted Rice Yield (ridge_trend)

Fig 8

Interpretation: Fig. 8 shows predictions closely aligned with actual yields, particularly between 1.0 and 4.0 t/ha, with greater dispersion at extreme values.

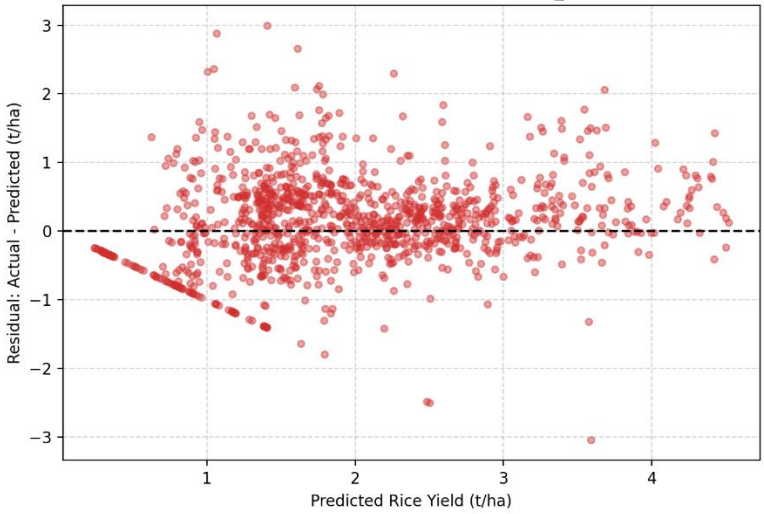


Fig 9: Locked Test Residual Plot (ridge_trend)

Fig 9

Interpretation: Fig. 9 shows largely homoscedastic residuals centered around zero, with negative errors from complete crop failures and positive outliers from high-performing districts exceeding trend projections.

9) Qualitative Error Analysis

The five largest absolute prediction errors on the locked test set were investigated from agronomic and socio-economic perspectives:

Row ID	Location & Year	Actual (t/ha)	Predicted (t/ha)	Absolute Error	Agronomic Cause & Policy Consequence	Practical Mitigation
D5_2024_2015	Ramgadh, Jharkhand (2015)	0.5550	3.5908	3.0358	Late August monsoon deficit and unlined tubewell depletion. Over-prediction risks under-allocating buffer grains.	Ingest near-real-time satellite NDVI and root-zone soil moisture telemetry during grain-filling.
D5_961_2014	Arwal, Bihar (2014)	4.3970	1.4045	2.9925	Solar lift-irrigation expansion along Son river basin and adoption of flood-tolerant seeds (Swarna-Sub1). Under-prediction causes	Incorporate annual agricultural credit, certified seed distribution, and energized tubewell densities.

Row ID	Location & Year	Actual (t/ha)	Predicted (t/ha)	Absolute Error	Agronomic Cause & Policy Consequence	Practical Mitigation
					warehouse bottlenecks.	
D5_37_2015	Sehore, Madhya Pradesh (2015)	3.9450	1.0634	2.8816	Conversion of black cotton soils to high-yield basmati via subsidized pressurized drip irrigation. Inaccurate supply quotas cause fertilizer shortages.	Partition spatial units by agro-ecological sub-zones rather than political district boundaries.
D5_956_2015	Lakhisarai, Bihar (2015)	4.2700	1.6110	2.6590	Delayed monsoon offset by emergency diesel subsidies and desilted traditional harvesting reservoirs (Ahar-Pyne). Inaccurate revenue projections delay market	Integrate state emergency relief announcements and surface water reservoir storage indices.

Row ID	Location & Year	Actual (t/ha)	Predicted (t/ha)	Absolute Error	Agronomic Cause & Policy Consequence	Practical Mitigation
					tariff collections.	
D5_2060_2015	Alipurduar, West Bengal (2015)	0.0000	2.5008	2.5008	Administrative bifurcation: Alipurduar was carved out of Jalpaiguri in 2014, leading to unrecorded returns entered as zero. Distorts PMFBY insurance baselines.	Implement automated boundary-harmonization scripts to link parent and bifurcated district registries.

10) Responsible Analytics

Ethical Governance and Data Rights

Agricultural predictive modeling directly influences public policy, farm credit ratings, and disaster relief allocations. The data utilized in this study represents aggregate district-level agricultural panel statistics, devoid of individually identifiable farmer information, geolocated land parcel boundaries, or proprietary financial records. If this pipeline is extended to farm-level decision support, explicit informed consent frameworks, differential privacy mechanisms, and spatial coordinate coarsening

must be implemented to protect individual landholders from predatory land speculation and discriminatory credit profiling.

Bias, Structural Disparities, and Agronomic Representation

The panel data exhibits geographic and historical imbalances. Districts in states with well-developed irrigation infrastructure (such as Punjab, Haryana, and coastal Andhra Pradesh) feature consistent historical reporting, extensive canal systems, and high baseline yields. In contrast, rain-fed districts in central and eastern India exhibit wider yield volatility, missing seasonal observations, and higher susceptibility to climatic shocks. A naive predictive model optimized for national aggregate loss functions risks prioritizing well-represented irrigated tracts while underperforming in marginalized, climate-vulnerable communities. Predictive errors must not be interpreted as differences in regional farmer competence; they reflect disparities in water infrastructure, rural capital access, and administrative data collection.

Failure Mode Risks and Human-in-the-Loop Safeguards

Yield predictions represent probabilistic statistical estimates, not agronomic facts. Relying solely on model outputs for central grain procurement or credit underwriting creates severe systemic risks:

- **Risk of Underprediction:** Induces artificial panic, prompting excessive commercial grain imports, depressing domestic farm-gate prices, and misdirecting subsidized fertilizer shipments away from high-performing tracts.
- **Risk of Overprediction:** Conceals impending localized food deficits, delaying the mobilization of state buffer stocks and emergency relief to famine-threatened rural populations.

Consequently, automated model forecasts must never directly trigger resource allocation decisions without independent field-level verification. Human oversight by local agronomists, regional extension officers, and community farmer organizations is

essential to validate model predictions against actual crop phenology, pest incidence, and weather conditions.

Limitations of Crop Recommendation Classifiers

The companion crop recommendation classification pipeline achieves high statistical accuracy (over 99%) on Dataset D3. However, this high performance must not be misinterpreted as an operational farming recommendation engine. The underlying dataset is synthetically balanced (exactly 100 samples across 22 crops) and relies solely on seven static soil and meteorological attributes. Real-world crop selection requires evaluating market price volatility, local transport logistics, capital requirements, seed variety availability, labor supply, and cultural dietary preferences. Recommending a crop based solely on physical soil compatibility without considering economic viability could encourage farmers to cultivate unmarketable produce, leading to severe financial distress.

11) Results and Discussion

Synthesis of Regression Findings

The prospective empirical findings demonstrate the necessity of aligning model inductive bias with non-stationary temporal dynamics. Across 22 training years (1990–2011), the Ridge Linear Trend model proved decisively superior to non-linear tree ensembles on both the prospective validation split (MAE: 0.4770 t/ha) and the locked test set (MAE: 0.4957 t/ha, RMSE: 0.6630 t/ha, $R^2 = 0.6889$), maintaining sub-millisecond inference latency (0.0065 s across 1,114 queries).

Tree ensembles were structurally constrained by their bounded leaf predictions, systematically underpredicting yields as national averages advanced secularly. Feature ablation confirmed that granular district identity is critical, with district omission causing a 39.79% degradation in predictive precision. Rolling-origin evaluations across 2009–2013 further verified that Ridge consistently maintained the lowest error across varying climate regimes.

Synthesis of Guided Crop Recommendation Classification Findings

The guided classification extension benchmarked environmental and soil chemical suitability across 22 crops in Dataset D3:

Model Architecture	Partition	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Status
Majority Baseline	Validation (N=440)	4.55%	0.0021	0.0455	0.0040	Benchmark
Multinomial Logistic Regression	Validation (N=440)	97.27%	0.9730	0.9727	0.9726	Evaluated
Random Forest Classifier	Validation (N=440)	**99.55%**	**0.9957**	**0.9955**	**0.9955**	**Selected**
Majority Baseline	Locked Test (N=440)	4.55%	0.0021	0.0455	0.0040	Benchmark
Random Forest Classifier	Locked Test (N=440)	**99.32%**	**0.9941**	**0.9932**	**0.9931**	**Final Test**

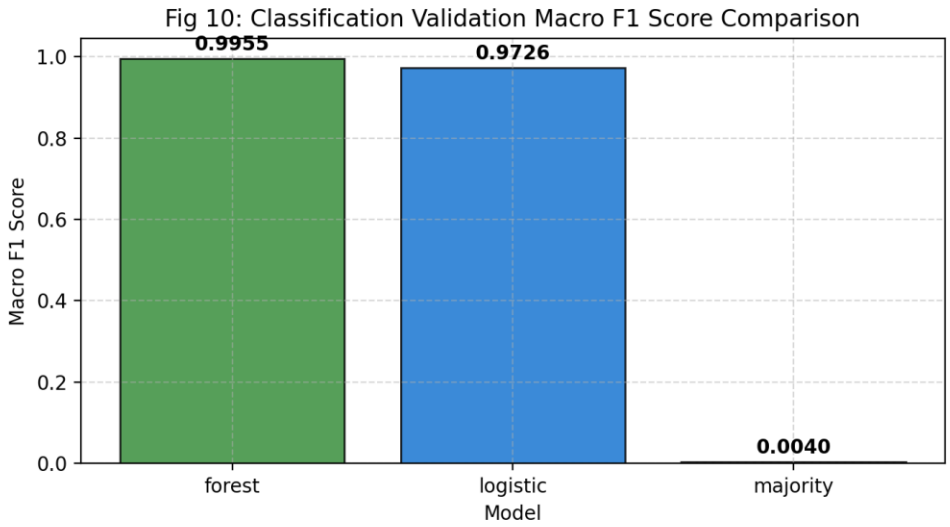


Fig 10

Interpretation: Fig. 10 shows strong classification performance, with Random Forest achieving a Macro F1 of 0.9955 and Logistic Regression 0.9726.

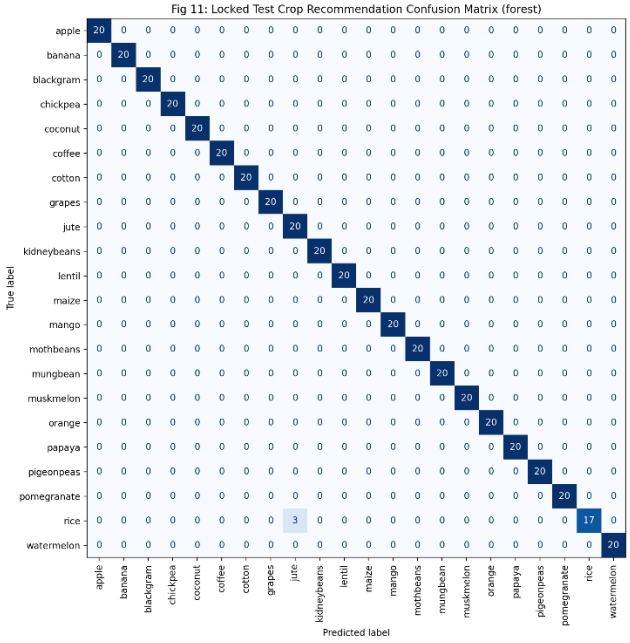


Fig 11

Interpretation: Fig. 11 shows Random Forest achieving 99.32% test accuracy, with only three misclassifications, all occurring between mothbeans and mungbean due to overlapping nutrient profiles.

12) Conclusion

This investigation demonstrated the construction, prospective validation, and diagnostic auditing of an end-to-end agricultural analytics pipeline. By enforcing pre-harvest feature whitelisting and strict chronological partitioning, the study established that agricultural yield forecasting requires models capable of extrapolating non-stationary secular trends.

The parametric Ridge Linear Trend model, combining district-level fixed effects with an annual linear slope, achieved superior prospective accuracy (Test MAE: 0.4957 t/ha, $R^2 = 0.6889$), decisively outperforming complex non-linear ensembles. Spatial ablation proved district identity essential, accounting for a 39.79% performance contribution. Diagnostic analysis of residual outliers identified localized monsoon shocks, accelerated solar irrigation, and administrative bifurcations as primary error drivers, emphasizing the necessity of satellite telemetry and human-in-the-loop agronomic oversight in real-world deployment.

Appendices

Appendix A: Environment Specifications

Conducted on Windows 11 Education (64-bit) with Python 3.14.0a2 and 16 GB RAM. Key libraries pinned in versions.json include scikit-learn 1.8.0, pandas 2.3.3, numpy 2.4.2, and matplotlib 3.10.8. A deterministic seed (SEED = 42) was enforced across all splits and estimators.

Appendix B: Model Configuration

Regression features were preprocessed via a ColumnTransformer (one-hot encoding for categorical variables; median imputation and standard scaling for harvest year). Evaluated models include a median baseline, Ridge regression (alpha = 1.0), a Decision Tree (max depth 6), and Random Forest ensembles (depths 6 and 12, 60

trees). The classification pipeline scaled soil and weather features, evaluating majority baseline, multinomial logistic regression, and random forest (60 trees) classifiers.

Appendix C: Key Concepts for Viva Voce

The prediction contract enforces strict pre-harvest inputs (state, district, season, year) to forecast rice yield (t/ha), excluding post-harvest production and area to prevent mathematical leakage. Chronological partitioning avoids temporal lookahead bias inherent in random splits. Ridge outpredicted tree ensembles because its linear year coefficient extrapolates upward secular trends, whereas tree leaves are bounded by training thresholds. For MAE, the median baseline minimizes expected L1 loss. Negative R^2 signifies predictive accuracy worse than the test mean under distribution drift. MAPE is excluded due to undefined zero-yield cases. Finally, a TEST_LOCK protocol guarantees single-pass evaluation, preventing test snooping.

Appendix D: Academic Integrity and Assistance Disclosure

Authored by Raghav Mritunajaya KS (Reg. No: 23MID0078) in compliance with VIT SCOPE academic integrity standards. AI assistance was restricted to code scaffolding, linting, and editing; all analytical pipelines, modeling, and diagnostics represent original work. Public datasets D5 (ICRISAT, DOI: 10.17632/ywp3y5j9vv.1) and D3 (Atharva Ingle) were acquired and cited under open-science terms.